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The relevance of media sentiment for small and large scale bitcoin investors

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ABSTRACT

We provide a novel perspective on the bitcoin market, investigating determinants of investor positions and their response to public information proxied by media sentiment indicators. We distinguish between investors by size and observe their respective behaviour concerning incoming information. We find that price dynamics and media sentiment lead to different decisions depending on the bitcoin portfolio size. Retail investors react strongly to incoming public information and media narratives, with their decisions strongly influenced by sentiment and media attention. Contrary to this, the response of large-scale investors to such information is much weaker because they arguably have different, non-public information and divergent investment objectives.

1. Introduction

Bitcoin[1] is growing as an asset and is increasingly attracting interest among researchers and investors, both from a theoretical and an empirical perspective (Garratt and Wallace, 2018; Hendrickson et al., 2016). Given the lack of fundamental models for the price of bitcoin, expectations and mood swings are widely considered to be important drivers of bitcoin prices. A broad strand of the literature has assessed potential determinants of bitcoin prices and price volatility and the link to other markets such as gold and stocks (Shahzad et al., 2019; Gkillas et al., 2020). Naturally, news in general and market sentiment in particular have emerged as a potential driver of bitcoin price dynamics (Akyildirim et al., 2021; Corbet et al., 2020a,b).

Recently, more and more institutional investors have expressed interest in Bitcoin. Despite the fact that various types of investors are getting exposure to bitcoin, the literature has mostly focused on price and price volatility but is notably silent when it comes to the behaviour of investors in the bitcoin market and the determinants of their investment decisions. It is not only important to understand the behaviour of short-term vs long-term investors but also the difference between small investors (retail) and large investors (institutional). The latter distinction is of particular importance given that the bitcoin market has attracted particular attention due to the discussion of whether a small number of large-scale investors can affect or manipulate the bitcoin market (Alexander and Heck, 2020; Auer et al., 2022; Fratrič et al., 2022).

So far, the existing literature often refers to investor behaviour when analysing hedge and safe haven properties based on price dynamics. However, this does not provide any information on the question of how different types of investors react to market

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[1] "Bitcoin" is capitalized when referring to the network and lower-cased when referring to the currency or price.

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developments, a topic which is becoming increasingly relevant for researchers, regulators and investors. While the literature on investor behaviour by investor type in the traditional financial sector is relatively scarce, it is almost non-existent for Bitcoin given the non-availability of adequate data. However, gaining insights into who is accumulating or selling bitcoin can provide important information on investor behaviour and the development of bitcoin as an asset.

Against this background, this paper relies on a unique data source by Glassnode extracting information from the Bitcoin blockchain to shed light on the behaviour of investors in the bitcoin market. Specifically, we investigate the behaviour of investors with different investment sizes in response to fluctuations in bitcoin prices and sentiment. Our study pays special attention to the behaviour of small versus large entities concerning their use of public information. To achieve this, we leverage the nature of the Bitcoin network's public ledger and use entity data to identify investor groups based on their bitcoin holdings. We analyse the response of the corresponding bitcoin positions of entities to media sentiment and price dynamics as proxies of public information about Bitcoin. Throughout our analysis, media sentiment can be distinguished between two variables: the sentiment, which reflects whether news are positive or negative, and media attention, which reflects the extent of mentions. This empirical setup accounts for the argument that media narratives affect economic behaviour (Shiller, 2017) and that the bitcoin market is strongly affected by mood swings. The data on sentiment and media attention we adopt covers a broad set of media content. It has been used by Michaelides et al. (2019), among others, as a proxy for public information on the currency market to assess the behaviour of currencies when private information becomes public.

We are the first to analyse investors' behaviour based on their bitcoin holdings using entity data at the daily level. We provide novel insights into the dynamics of the bitcoin market and add a new perspective to the existing literature. On the one hand, we contribute to the existing literature by examining whether the behaviour of different investor groups varies. It is commonly assumed that large entities are primarily controlled by institutional investors or "smart money", which may react differently to price shocks or sentiment shifts than smaller entities. Retail investors, who typically hold smaller positions and have less experience investing, may be more sensitive to price and sentiment shocks and more likely to panic sell during periods of extreme fear and price drops. In contrast, institutional investors may view such periods as an opportunity to accumulate bitcoin. As a second contribution, we analyse whether diverging reactions across entities may be due to differing responses to public information.

The remaining paper is structured as follows. The following section presents an elaborate overview of related literature. A detailed data description will be given in Section 3. Section 4 describes the overall methodology. Section 5 provides empirical findings with an in-depth discussion and an evaluation of the hypotheses. Section 6 concludes the paper.

2. Literature review

Research on Bitcoin is relatively new but growing quickly. This section starts by briefly discussing previous research related to our research question but with a focus on price or price volatility rather than investment positions. As a second step, we focus on studies which deal with investor behaviour in the bitcoin market before we turn to the existing evidence of the role of sentiment in price discovery.

A significant part of research deals with the nature of bitcoin, whether it is digital gold, a digital currency or just a speculative asset (Wu et al., 2019; Shahzad et al., 2020). Another strand of the literature analyses how policy uncertainty affects bitcoin returns, with studies frequently relying on text-based economic policy uncertainty measures. These studies are relevant to the investigation because they are based on the argument that investor behaviour in the bitcoin market is affected by uncertainty, which propagates into price dynamics. Several authors like Wang et al. (2019), Fang et al. (2019), Wu et al. (2021) or Demir et al. (2018) focus on this topic. They find that bitcoin acts as a hedge or weak hedge against uncertainty, which can mainly be observed for extreme uncertainty shocks. The aforementioned studies analyse the effects of economic policy uncertainty at the global level. Wüstenfeld and Geldner (2021) have conducted a similar study for bitcoin at the national currency level for eight countries. Their results suggest that it depends on the studied currency or country whether bitcoin acts as a hedge against uncertainty. Based on different proxies, papers by Aysan et al. (2019) and Al Mamun et al. (2020) analyse the effects of geopolitical risks on bitcoin returns and volatility, relying on the same argument. Recent work also has used textual analysis to assess uncertainty surrounding the crypto market (Lucey et al., 2022).

While such studies focus on the response of the bitcoin market to uncertainty directly, a second strand of the literature has investigated the relationship between other market indices, various assets and bitcoin directly. The Bitcoin market has, for example, been linked to oil and gold prices, exchange rates, stock and exchange indices and volatility indices such as the VIX (see Gkillas et al., 2020; Shahzad et al., 2019; or Guesmi et al., 2019). Despite the growing body of research on Bitcoin, one area that remains relatively underexplored is the role and behaviour of different types of investors in response to changing market conditions. While some authors such as Lammer et al. (2020) and Bourghelle et al. (2022) have attempted to tackle this topic, no independent discussion has yet developed around these relevant issues. For instance, Lammer et al. (2020) try to identify who invests in bitcoin using indirect investment data, while Diaconaşu et al. (2022) study investor behaviour based solely on returns and trading volume. Chan et al. (2020) analyse investor data from an Asian medium-sized cryptocurrency exchange and distinguish between institutional and individual investors to investigate their trading behaviour. Jalan et al. (2023) identify a positive and statistically significant effect of trust on interest in cryptocurrencies. However, the existing literature only approximately distinguishes between different types of investors and draws conclusions about investors broadly, without differentiating between small, medium, and large investors. The role of large investors is only discussed indirectly and with a primary focus on small investors.

Our work also relates to the role of news for the bitcoin market. Both macroeconomic news and the role of monetary policy announcements have been proposed as potential drivers of bitcoin prices without an explicit focus on investors (Corbet et al.,

Table 1
Entity categories by bitcoin holdings.

	Size of smaller entities		Size of bigger entities
1.	0–0.01 BTC (0–0.001 BTC + 0.001–0.01 BTC)[a]	**5.**	100–10k BTC (100–1k BTC + 1k–10k BTC)[a]
2.	0.01–0.1 BTC	**6.**	10k–100k BTC
3.	0.1–1 BTC (1–10 BTC + 10–100 BTC)[a]	**7.**	>100k BTC

[a] Note: Original entity categories in parentheses if aggregated.

2020a,b). The results show, for example, that an increase in the percentage of negative news coincides with higher bitcoin returns. Following the pioneering work of Baker and Wurgler (2007), who focus on stock prices, sentiment analysis has emerged as a rapidly growing area of research analysing investor behaviour in financial markets. Studies have also used various sentiment variables to proxy investor behaviour in the bitcoin market. For instance, Chen et al. (2020) examined the impact of fear sentiment on bitcoin price dynamics at the onset of the COVID-19 pandemic. They used Google Trends data and the VIX index to measure investor sentiment and expectations. Similarly, Guégan and Renault (2021) found a significant relationship between investor sentiment and bitcoin returns. They extracted sentiment from social media messages and used high-frequency bitcoin trading data. Bourghelle et al. (2022) investigated the bilateral relations between bitcoin volatility and investor emotions, using bitcoin price data and the Fear & Greed sentiment index to proxy for bitcoin investor behaviour. Burggraf et al. (2021) find that investor sentiment has strong predictive power for bitcoin's price by constructing a sentiment index and applying various quantitative methodologies such as the transfer entropy model, threshold regressions and linear estimations. Overall, the existing evidence points out that sentiment is of potential relevance for the bitcoin market and investors, illustrating the need to account for information in sentiment when analysing investment behaviour. From this perspective, we explore the gap in the literature, which arises from the fact that the role of sentiment dynamics for disaggregated investor data has not been analysed yet.

3. Data and preliminary analysis

3.1. Entity data

To identify the buying and selling behaviour of entities on the Bitcoin network, we use the entity data set by the data provider Glassnode, which runs from January 2017 to December 2021. To the best of our knowledge, we are the first to use that data in economic research. In the following, we will summarise why the underlying information provides an adequate proxy for investment behaviour in the bitcoin market.

Every transaction made within the Bitcoin network since its inception is recorded on the blockchain. Despite misconceptions, the network's transactions are not fully anonymous but rather pseudo-anonymous. This means that the sending and receiving history of all public addresses in the network can be observed, enabling the user to track and observe all activity that has ever been conducted on-chain. This feature presents new opportunities for researchers to obtain and analyse data at a relatively low threshold. On-chain data is a unique data source for financial assets. However, a potential issue with raw on-chain data is that a single user may have multiple public addresses, making it difficult to accurately analyse the behaviour of small and large holders using public addresses and their associated transaction history.

Glassnode addresses this issue with their entity data "by applying a combination of industry-standard heuristics, proprietary clustering algorithms and advanced data science methods on top of raw on-chain data" (Glassnode (2020)). In other words, they attribute multiple public addresses to single entities. A major advantage of entity data over more easily accessible data, such as trading volume, is that it allows a differentiated analysis of how investors adjust their bitcoin holdings over time. We have data on entity holdings of ten different sizes, but for the later analysis, we have aggregated them into seven entities. The entity categories are shown in Table 1. The data is primarily aggregated for two reasons: First, using the more detailed entities in the estimations does not add many more insights. Thus, reducing the number of entities facilitates the interpretation thereof. Second, broader entity categories reduce the number of entities that switch between categories. For example, suppose an entity holding just below ten bitcoins adds a few more bitcoins to its holdings and increases its holdings above the ten bitcoin threshold. In that case, it would move into the 10 – 100 bitcoin category. By using broader entity categories, we minimise the number of entities that switch between categories.

It is important to note that these are entities and not single individuals. An entity can be an individual, but it can also be an entity that holds the funds of multiple individuals. For instance, an entity holding funds of multiple users can be an investment fund or an exchange. Entities holding more than 100k bitcoin, in particular, are mostly exchanges holding the funds of millions of users. Similarly, entities holding 10k – 100k bitcoin may also be subject to that. Net growth in these two categories does not necessarily mean people are accumulating bitcoin. On the contrary, it could mean that individuals deposit their bitcoin onto exchanges to sell. For the sake of completeness, however, we include them in our estimations. Entities smaller than 10k bitcoin should not be affected by this and thus are suitable for our analysis to differentiate between small and large holders.

Before delving into the data, it is important to understand why we start our analysis in 2017. Although Bitcoin has been around for over a decade, it is still a relatively young asset in terms of market depth and adoption. Particularly in the early years, data

Table 2
Entity categories: Lower and upper bounds of amount of entities.

	Entities	Lower bound	Upper bound
		Mean over observation period	
1.	0–0.01 BTC	1.962.661	X
2.	0.01–0.1 BTC	1.764.715	17.647.150
2.	0.1–1 BTC	483.666	4.836.663
4.	1–100 BTC	44.503	4.450.301
5.	100–10k BTC	716	71.550
6.	10k–100k BTC	20	200
7.	>100k BTC	X	35
	Number of entities	**4.3 m**	**29 m**[a]
		Date of observation: 01/01/2017	
1.	0–0.01 BTC	597.537	X
2.	0.01–0.1 BTC	430.200	4.302.000
2.	0.1–1 BTC	220.465	2.204.651
4.	1–100 BTC	37.551	3.755.056
5.	100–10k BTC	692	69.250
6.	10k–100k BTC	29	295
7.	>100k BTC	X	28
	Number of entities	**1.3 m**	**10.9 m**[a]
		Date of observation:12/31/2021	
1.	0–0.01 BTC	3.742.214	X
2.	0.01–0.1 BTC	1.793.479	17.934.794
2.	0.1–1 BTC	736.555	7.365.554
4.	1–100 BTC	47.102	4.710.186
5.	100–10k BTC	771	77.136
6.	10k–100k BTC	18	183
7.	>100k BTC	X	37
	Number of entities	**6.3 m**	**33.8 m**[a]

[a] Note: For the smallest entity category (0 – 0.01 BTC) we cannot calculate an upper bound. Due to this, for the calculation of the aggregated upper bound number of entities we have used the lower bound calculation of that category. So in reality the upper bound numbers will be higher. Similarly the upper bound numbers for the largest entity category have been used in the lower bound calculations.

on Bitcoin may be unreliable. It may potentially be influenced by only a few participants or unusual market movements. As a result, the early days are unsuitable for analysing entity behaviour. Since 2017, however, there has been a significant increase in cryptocurrency and Bitcoin adoption, as evidenced by real trading volume data from messari.io. For instance, the average daily real trading volume for bitcoin surged from $31 million in 2016 to $560 million in 2017.[2] This represents an 18-fold increase in trading activity. Furthermore, while in 2016, on average, each day 66k new entities were created, in 2017, that number increased to 103k, a 57% increase compared to the previous year. These figures demonstrate significant growth in the adoption and interest in Bitcoin, making data from 2017 onward more reliable and representative of the market. Therefore, we use data from 2017 onward for our analysis, covering the period from January 1, 2017, to December 31, 2021.

The increase in adoption can also be observed by examining our dependent variables. Although we cannot determine the exact number of entities in each category, we can estimate the upper and lower bounds of the number of entities for each category based on the provided data. Since the possible number of entities in each category changes over time, we present the upper and lower bounds for the mean of the studied period, as well as the start and end points of the studied period, in Table 2. This gives the reader a better understanding of how entities change over time[3]. At the beginning of 2017, the Bitcoin network had at least 1.3 million entities. By the end of 2021, this number had increased to 6.3 million, representing an almost five-fold increase. However, based on the available data, this number only represents the absolute lower bound. The upper bound increased almost three-fold, from 10.9 million to 33.8 million entities. It is important to note that the upper bound is likely significantly higher, as we cannot calculate an upper bound for the smallest entity category (in theory, it is only limited by the divisibility of bitcoin into sub-units). The number of small entities is in the millions, while there are only a few hundred, if not fewer, large entities. For example, at the beginning of the study period, there were a maximum of 295 entities holding 10k – 100k bitcoin, but it is more likely that the real number is closer to the lower bound of 29 entities.

[2] According to a report by Bitwise (2019) compiled for the SEC, nearly 95% of reported trade volume can be considered fake volume (attributable to, e.g. wash trading). Other than trade volume data from, e.g. Coinmarketcap.com, the trade volume data from messari.io has the advantage of only considering trading volume from exchanges that are not susceptible to wash trading. Therefore, we consider the messari data to be more representative of trading activity at the global level.

[3] The descriptive statistics of the entity data can be found in Table 4.

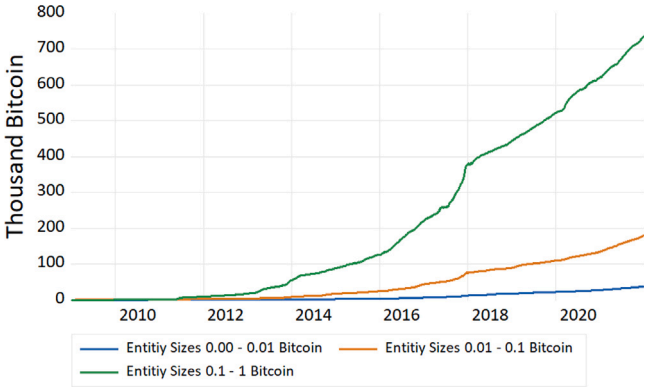


Fig. 1. Small entities cumulative bitcoin holdings; 01/03/2009–12/31/2021.

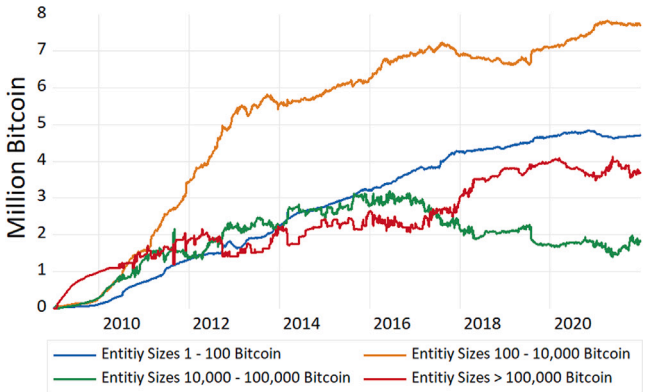


Fig. 2. Large entities cumulative bitcoin holdings; 01/03/2009–12/31/2021.

Fig. 1 shows the cumulative holdings of the three smallest entities over time. It is evident that their holdings are trending upward. Minor fluctuations and slight declines can be observed, but smaller investors are, on the aggregate, either becoming long-term holders and growing in numbers and/or continuing to accumulate bitcoin over time. The entity category with holdings of 0.1 – 1 bitcoin (the largest entity category in the figure) also has the highest holdings compared to the other two. For instance, at the end of the observation period 12/31/2021, entities holding 0.1 – 1 bitcoin held 736,555 bitcoin, while entities holding 0.00 – 0.01 bitcoin on the aggregate held only 37,422 bitcoin. The four largest entities' bitcoin holdings are depicted in Fig. 2. These entities exhibit more volatility in their bitcoin holdings over time than the three smallest entities. In contrast to the small entities, the data shows greater variation, including prolonged periods of declines in holdings. In fact, the aggregated bitcoin holdings of entities with a balance of 10k – 100k bitcoin have been in a macro downward trend since 2016. With the increase in bitcoin's price, it is not surprising that large entities are selling some of their bitcoin holdings. Although the majority of bitcoin are still held by larger entities, the data suggests that bitcoin are being distributed across the network over time, disproving the common narrative that bitcoin are increasingly concentrated in the hands of a few large players. The cumulative bitcoin holdings show differences and commonalities between the various entity sizes. Looking at the percentage changes of entity holdings provides more insights into the volatility of these holdings and how they change over time. They are displayed in Fig. 3 for the studied period 2017–2021.

The data shows that changes in bitcoin holdings occurred mainly during the 2017 bull market, with some fluctuations in 2018 depending on the entity category. The smallest entity category experienced particularly positive changes, while for other small entities, it was a mix of increases and decreases in holdings. After 2017 and partially 2018, changes in bitcoin holdings of these smaller entities became less pronounced, except for a slight uptick in activity during 2021. This pattern is not surprising, as the bear market of 2018 would naturally result in decreased activity among smaller entities. However, it is somewhat unexpected that we did not observe a significant increase in activity during the recent bull market, unlike in 2017. This suggests that there may not have been a retail mania this time.

The two largest entity categories exhibit the most volatile changes in bitcoin holdings, with relatively similar patterns that appear unnatural compared to the other entities. As previously mentioned, the largest entities could be exchanges or funds holding bitcoin for multiple individuals, and the similar pattern of both entities suggests that this might be the case for them. In contrast, the third largest entity category displays a more typical behaviour with one or two larger spikes. Compared to small entities, the behaviour of larger entities does not appear to be heavily influenced by cycles. Although percentage changes were more significant in 2017 than

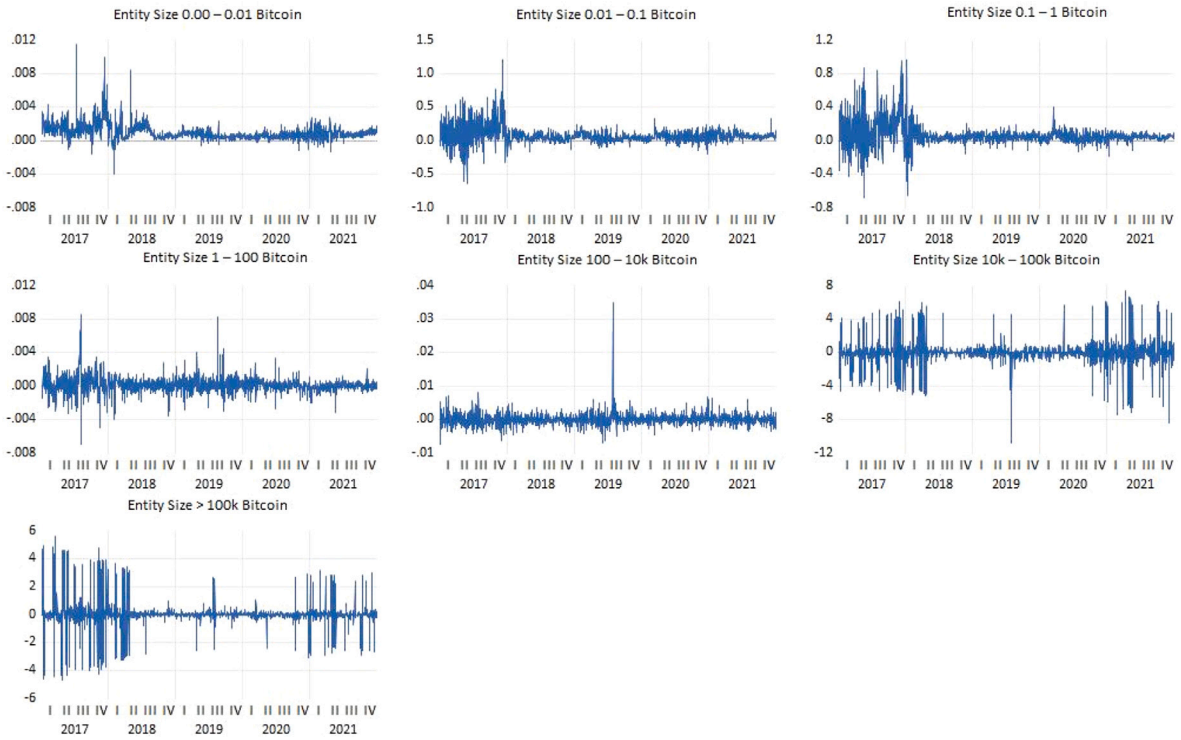


Fig. 3. Percentage change all entities cumulative bitcoin holdings; 01/01/2017–12/31/2021.

Table 3
Entity correlations 2017–2021.

Entities	0–0.01	0.01–0.1	0.1–1	1–100	100–10k	10k–100k	>100k
0–0.01	1.0000						
0.01–0.1	0.9975	1.0000					
0.1–1	0.9911	0.9923	1.0000				
1–100	0.8713	0.8650	0.9167	1.0000			
100–10k	0.7433	0.7505	0.7086	0.5039	1.0000		
10k–100k	−0.8367	−0.8378	−0.8773	−0.9044	−0.5480	1.0000	
>100k	0.7653	0.7505	0.8040	0.8833	0.2287	−0.8865	1.0000

Note: The table reports the correlations between the bitcoin entities for the period 01/01/2017–12/31/2021.

in the following years, they do not stand out as much. Overall, there are notable differences in patterns when comparing smaller entities to larger ones. This suggests that small investors may behave differently than large investors. Therefore, it is worthwhile to study different investor types, as they might react differently to price and sentiment shocks.

This difference between small and large entities is also evident in the correlations shown in Table 3. The three smallest entity categories are highly correlated, with coefficients above 0.99. Small entities are not as highly correlated with larger entities. Across all entities, the correlations are generally positive, with the exception of entities holding 10k – 100k bitcoin, which are negatively correlated with all other entities.

Entity data is a new concept meaning that investors have yet to use this data to make investment decisions. This is beneficial for our analysis as it eliminates reverse causality problems. The size of entities is not published in real-time, which eliminates the possibility of direct effects on bitcoin price or sentiment by investors watching the data. However, individual investors may still affect prices or volatility. The sentiment variables we use should not directly be affected by shifts in entity holdings, but there may be an indirect effect on sentiment via price. Nonetheless, Granger causality tests and regressions conducted within our empirical analysis show no evidence of reverse causality. This does not exclude the possibility that individual investors affect price or market sentiment but confirms that such a pattern is not detected over the whole time span at the aggregated level.

3.2. Sentiment data

We consider the Thomson Reuters MarketPsych Indices (TRMI) to analyse whether sentiment factors influence bitcoin investment decisions, . MarketPsych uses lexical analysis to extract sentiment indices by evaluating various sources (social media platforms,

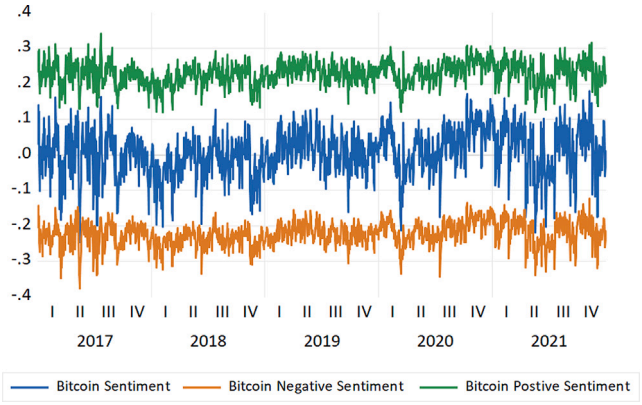


Fig. 4. Bitcoin sentiment; 01/01/2017–12/31/2021.

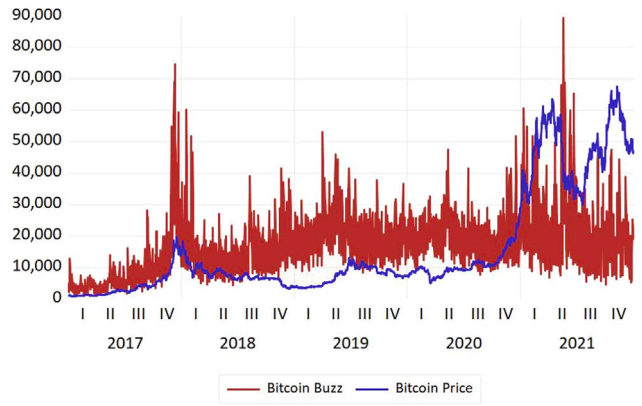


Fig. 5. Bitcoin price development and buzz index; 01/01/2017–12/31/2021.

such as Twitter, and newspaper articles) every minute, adding up to over two million daily posts and news articles being covered in the data. Because our dependent variables are reported daily, we rely on the reported daily TRMIs. Each sentiment index is the weighting of words and phrases associated with a particular type of sentiment (for example, fear or euphoria) in the overall text of the combined news and social media content. The daily index is a simple average of the last 24 h or 1440 min of information (Peterson, 2013). Thus, the TRMI represents an unmatched collection of premium news and a wide range of social media platforms, making it currently the most comprehensive sentiment index available. The TRMI indices scores are either bipolar (−1 to 1) or unipolar (0 to 1). Data by MarketPsych has been adopted as a proxy for public information on financial markets since it is available in real-time and reflects various perceptions of financial markets in the media. Therefore, we argue that such indicators, next to price dynamics, reflect the most important sources of public information and also capture media narratives about Bitcoin (Michaelides et al., 2019).

We mainly use the general sentiment index in our analysis which is bipolar, describing the sum of the overall positive and negative references regarding Bitcoin.[4] The bipolar sentiment index and the unipolar positive and negative indices are displayed in Fig. 4. In general, all three indices exhibit significant volatility, with periods of more extreme movements up and down. However, due to the nature of the variables, it is difficult to identify specific events by simply examining the figure. We also include the buzz index, which is unique compared to the other indices. It ranges from 0 to theoretically infinity because it is based on the frequency of news stories related to the filtered subject (in this case, Bitcoin) and represents our proxy for media attention. Due to this, we can also observe trends graphically which for the other three sentiment variables is due to the way that they are scaled not that easy. The descriptive statistics of the independent Bitcoin variables are presented in Table 5.

Fig. 5 displays the daily Bitcoin buzz and bitcoin price development between 2017 and 2021. It is apparent that over time sharp declines in sentiment are usually accompanied by sharp price declines. Although sentiment and price often move in the same direction, their magnitudes of the shift are not always aligned. At times, both variables move in opposing directions. Therefore, including both variables in our analysis is warranted and can give additional insights.

[4] To look at the negative and positive sentiment influences separately, we also use the unipolar indexes in each case in our analysis, elaborated in the robustness tests.

Table 4
Descriptive statistics of the bitcoin entities.

Entity size BTC	0.00–0.01	0.01–0.1	0.1–1	1–100	100–10k	10k–100k	>100k
Mean	19,626.61	103,024.60	483,666.30	4,450,301.00	7,155,044.00	1,996,829.00	3,482,869.00
Median	19,860.33	101,397.60	480,383.10	4,505,276.00	7,088,398.00	1,915,592.00	3,682,853.00
Maximum	37,422.14	179,347.90	736,555.40	4,846,123.00	7,826,578.00	3,021,693.00	4,119,827.00
Minimum	5,975.37	43,019.95	219,675.60	3,735,428.00	6,625,680.00	1,391,528.00	2,151,284.00
Std. Dev.	8,297.17	35,627.38	141,279.50	305,069.50	368,132.30	340,745.70	522,107.70
Skewness	0.17	0.22	−0.14	−0.83	0.48	0.90	−1.15
Kurtosis	2.12	2.25	2.10	2.68	1.96	3.36	3.03
Jarque–Bera	67.03	57.30	67.62	219.85	153.24	258.53	405.45
Probability	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sum	3.58E+07	1.88E+08	8.83E+08	8.13E+09	1.31E+10	3.65E+09	6.36E+09
Sum Sq. Dev.	1.26E+11	2.32E+12	3.64E+13	1.70E+14	2.47E+14	2.12E+14	4.97E+14
Observations	1826	1826	1826	1826	1826	1826	1826

Note: The table reports the descriptive statistics of the bitcoin entities for the period 01/01/2017–12/31/2021.

Table 5
Descriptive statistics of the independent bitcoin variables.

	BTC Price	BTC Sentiment	BTC S_positive	BTC S_negative	BTC Buzz
Mean	15,492.52	0.01	0.23	0.22	3.98
Median	8,635.89	0.01	0.23	0.22	3.87
Maximum	67,617.02	0.18	0.34	0.38	25.36
Minimum	784.28	−0.25	0.12	0.12	−1.41
Std. Dev.	17,002.96	0.06	0.03	0.03	3.23
Skewness	1.56	−0.44	−0.36	0.46	1.17
Kurtosis	4.04	3.44	3.30	3.75	6.37
Jarque–Bera	825.23	74.70	45.89	106.58	1280.31
Probability	0.00	0.00	0.00	0.00	0.00
Sum	2.83E+07	10.40	417.74	407.34	7263.71
Sum Sq. Dev.	5.28E+11	7.54	1.94	2.02	19,094.61
Observations	1826	1826	1826	1826	1826

Note: The table reports the descriptive statistics of the independent variables bitcoin price, Bitcoin sentiment, Bitcoin sentiment positive and negative and Bitcoin buzz for the period 01/01/2017–12/31/2021.

Table 6
Descriptive statistics of the independent Gold variables.

	Gold Price	Gold Sentiment	Gold S_positive	Gold S_negative	Gold Buzz
Mean	1,501.97	0.06	0.28	0.21	14,714.58
Median	1,411.40	0.07	0.28	0.21	14,468.98
Maximum	2,069.40	0.26	0.39	0.34	80,193.15
Minimum	1,162.00	−0.20	0.15	0.12	852.90
Std. Dev.	254.34	0.06	0.03	0.03	9,280.73
Skewness	0.40	−0.34	−0.15	0.46	0.83
Kurtosis	1.57	4.18	3.96	3.99	5.32
Jarque–Bera	140.99	141.79	77.70	139.15	618.44
Probability	0.00	0.00	0.00	0.00	0.00
Sum	1.89E+06	118.17	503.37	385.20	2.69E+07
Sum Sq. Dev.	8.14E+07	5.60	1.78	1.41	1.57E+11
Observations	1260	1826	1826	1826	1826

Note: The table reports the descriptive statistics of the independent variables Gold Price, Gold Sentiment, Gold Sentiment positive, Gold Sentiment negative and Gold Buzz for the period 01/01/2017–12/31/2021.

As control variables, we also include gold sentiment and gold prices since bitcoin is often referred to as digital gold. We also consider the economic policy uncertainty index (EPU) by Baker et al. (2016) and the VIX index as established measures of uncertainty which have been widely adopted when analysing hedge and safe haven properties. Finally, we include an extension of daily dummies for FED statements provided by Cieslak and Schrimpf (2019)and the effective federal funds rate to account for the monetary policy environment. The summary statistics of these variables can be found in Tables 6 and 7.

Table 7
Descriptive statistics of the control variables.

	EPU	VIX	FED statements	Interest rate
Mean	0.96	18.42	0.04	1.09
Median	0.13	16.28	0.00	1.16
Maximum	16.35	82.69	1.00	2.45
Minimum	−2.16	9.14	0.00	0.04
Std. Dev.	2.52	8.78	0.19	0.87
Skewness	2.27	2.69	4.93	0.09
Kurtosis	9.18	14.42	25.29	1.52
Jarque–Bera	4477.85	8364.06	45,200.54	169.27
Probability	0.00	0.00	0.00	0.00
Sum	1,750.55	2.32E+04	67.00	1,988.19
Sum Sq. Dev.	11,597.54	9.69E+04	64.54	1,390.94
Observations	1826	1259	1826	1826

Note: The table reports the descriptive statistics of the control variables EPU, VIX, Fed statements and Interest Rates for the period 01/01/2017–12/31/2021.

4. Methodology

Upon examining the bitcoin price development, the standard linear regression model assumption that the model parameters do not vary across observations is highly questionable.

Therefore, we are testing and estimating models that allow for structural changes. To address this, we apply linear breakpoint regressions, allowing for changes in the relationship between variables over time. The regime breakpoints may be known and specified a priori, or they may be estimated using the methods developed by Bai (1997) or Bai and Perron (1998). We also adopt robust standard errors and adopt different lag orders to account for autocorrelation and errors which are robust to heteroscedasticity. Finally, we conduct unit root tests which confirm that all error terms are stationary.

Our main specification is based on a multiple linear regression with m potential breakpoints and $m+1$ regimes:

$$y_t = X_t'\beta_t + \varepsilon_t,\ (t = T_{j-1}+1,\ ...,\ T_j), \tag{1}$$

with the convention that $T_0 = 1$ and $T_{m+1} = T+1$. y_t denotes the change of the respective entity group, X_t' denominates the vector of the regressors and β_t is the coefficient vector which is also subject to possible regime shifts. ε_t denotes an error term.

To select the dimension of the model, we apply the Bayesian Information Criterium (BIC), which according to Bai and Perron (1998, 2003) works well in most cases when breaks are present. After calculating the tests for all possible breakpoints, the sequence is selected as the configuration at which the BIC achieves its minimum. The breakpoints obtained in this fashion are a local minimum of the sum of squared residuals given the number of breakpoints but are not necessarily a global minimum.

Granger causality tests and estimates with sentiment and prices as independent variables confirm that entities should be considered as the left-hand side variable. We, therefore, employ the change of each entity group as the dependent variable. Given that we hypothesize that the effect of sentiment and prices differs across entities, we employ separate time series estimates for each entity group.

5. Empirical results

5.1. Main results

We have estimated two variations of breakpoint regressions. The first variant involves considering regressions with a priori fixed breakpoints, and for the second variant, breakpoints are estimated using the method developed by Bai (1997) and Bai and Perron (1998). All estimates are conducted with white robust standard errors to account for heteroskedasticity. We conduct entity-specific estimates and label the first three entities small, the fourth and fifth entity medium and the sixth and seventh entity large.

We start our assessment with the first breakpoint variant, where we choose January 1st, 2020, as the breakpoint as this date approximately marks the onset of the COVID-19 pandemic, which had a substantial impact on economies, financial markets, and societies worldwide, including the collapse of bitcoin' price by almost 50 percent.

The results of this regression can be found in Table 8 – Table 10, where we analyse each investor entity separately and examine the impact of our independent variables on each entity. The independent variables we consider here include the change in bitcoin price, Bitcoin sentiment, Bitcoin buzz, gold sentiment, and gold buzz, while EPU, Federal Reserve Bank (FED) communication, and Federal Funds effective interest rates are included as control variables. We used several transformations of the variables, which will be discussed in the next section.

The results clearly demonstrate that our empirical approach works much better for smaller entities. The adjusted R-Squared drops from 46% for the smallest entity to around 2% for the largest entity. At the same time, the coefficient estimates are much more significant for small and medium-sized entities. This is a key result since it establishes that public information only affects the

Table 8
Breakpoint regressions with one break, ΔEntity Groups 1, 2 and 3.

	ΔEntity Group 1 Size 0.00–0.01 Bitcoin		ΔEntity Group 2 Size 0.01–0.1 Bitcoin		ΔEntity Group 3 Size 0.1–1 Bitcoin	
Regime	1	2	1	2	1	2
Observations	1095	731	1095	731	1095	731
ΔEntity Group (−1)	**0.4779*****	**0.1832*****	**0.2801*****	**0.1015****	**0.1957*****	**0.3045*****
	(0.0629)	(0.0422)	(0.0453)	(0.0441)	(0.0618)	(0.0421)
ΔBTC Price	−0.0005	−0.0003	**0.0027***	**−0.0016****	**0.0058*****	**−0.0022****
	(0.0010)	(0.0005)	(0.0016)	(0.0007)	(0.0016)	(0.0009)
BTC SENTIMENT	**0.1853*****	0.0289	0.0489	**−0.0927*****	−0.1249	−0.0466
	(0.0554)	(0.0310)	(0.0999)	(0.0322)	(0.1120)	(0.0308)
BTC BUZZ	**0.0106*****	**0.0058*****	**0.0121*****	**0.0063*****	**0.0071****	−0.0001
	(0.0020)	(0.0007)	(0.0028)	(0.0008)	(0.0034)	(0.0009)
GOLD SENTIMENT	−0.0389	**0.0548***	0.0886	−0.0463	0.1066	**−0.1202*****
	(0.0545)	(0.0280)	(0.0736)	(0.0386)	(0.0918)	(0.0362)
GOLD BUZZ	−0.0011	**0.0046*****	**0.0140*****	**0.0063*****	**0.0164*****	**0.0059*****
	(0.0025)	(0.0010)	(0.0040)	(0.0012)	(0.0046)	(0.0013)
EPU	−0.0011	**−0.0024*****	−0.0008	**−0.0018*****	−0.0003	−0.0003
	(0.0024)	(0.0005)	(0.0032)	(0.0006)	(0.0040)	(0.0006)
FED STATEMENTS	0.0009	0.0062	**−0.0274****	**0.0321*****	0.0033	0.0130
	(0.0076)	(0.0087)	(0.0123)	(0.0106)	(0.0179)	(0.0102)
Interest Rate	**−0.0669*****	**−0.0171*****	**−0.0887*****	**−0.0215*****	**−0.0827*****	**−0.0090****
	(0.0080)	(0.0026)	(0.0099)	(0.0038)	(0.0119)	(0.0036)
R2	0.4607		0.2677		0.1881	

Note: The table reports the results of the least square regressions with one breakpoint of the entity groups 1, 2 and 3. The breaks are manually determined. Regime 1: 01/01/2017–12/31/2019. Regime 2: 01/01/2020–12/31/2021. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

Table 9
Breakpoint regressions with one break, ΔEntity Groups 4 and 5.

	ΔEntity Group 4 Size 1–100 Bitcoin		ΔEntity Group 5 Size 100–10k Bitcoin	
Regime	1	2	1	2
Observations	1095	731	1095	731
ΔEntity Group (−1)	0.0790	0.0089	−0.0297	−0.0168
	(0.0700)	(0.0612)	(0.0242)	(0.0420)
ΔBTC Price	**0.0043*****	**−0.0010***	−0.0009	0.0011
	(0.0011)	(0.0007)	(0.0014)	(0.0014)
BTC SENTIMENT	−0.0803	**−0.0875****	−0.1125	−0.0651
	(0.0668)	(0.0368)	(0.1053)	(0.0880)
BTC BUZZ	**−0.0071*****	**−0.0052*****	**−0.0087*****	0.0006
	(0.0019)	(0.0012)	(0.0030)	(0.0020)
GOLD SENTIMENT	**0.1430****	−0.0175	0.0989	0.0618
	(0.0599)	(0.0431)	(0.0938)	(0.0914)
GOLD BUZZ	**0.0130****	**0.0042*****	0.0071	0.0003
	(0.0029)	(0.0015)	(0.0049)	(0.0029)
EPU	0.0015	0.0000	0.0023	**0.0023***
	(0.0031)	(0.0006)	(0.0045)	(0.0013)
FED STATEMENTS	**−0.0265***	0.0018	**−0.0580***	**0.0564***
	(0.0161)	(0.0119)	(0.0310)	(0.0327)
Interest Rate	0.0031	0.0000	**0.0270***	0.0027
	(0.0075)	(0.0066)	(0.0141)	(0.0105)
R2	0.0761		0.0151	

Note: The table reports the results of the least square regressions with one breakpoint of the entity groups 4 and 5. The breaks are manually determined. Regime 1: 01/01/2017–12/31/2019. Regime 2: 01/01/2020–12/31/2021. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

behaviour of small and medium but not that of large entities. This finding is also robust to other break configurations or forms of nonlinearities.

Table 10
Breakpoint regressions with one break, ΔEntity Groups 6 and 7.

	ΔEntity Group 6 Size 10k–100k Bitcoin		ΔEntity Group 7 Size >100k Bitcoin	
Regime	1	2	1	2
Observations	1095	731	1095	731
ΔEntity Group (−1)	**−0.1290*****	**−0.1028***	**−0.1210****	−0.0160
	(0.0464)	(0.0622)	(0.0518)	(0.0667)
ΔBTC Price	0.0029	0.0100	−0.0117	−0.0014
	(0.0103)	(0.0171)	(0.0088)	(0.0072)
BTC SENTIMENT	−0.0231	1.1872	0.5191	−0.3198
	(0.7364)	(1.0631)	(0.6317)	(0.4207)
BTC BUZZ	0.0183	−0.0035	0.0227	0.0086
	(0.0240)	(0.0279)	(0.0192)	(0.0111)
GOLD SENTIMENT	−0.3860	0.0406	−0.2636	−0.0266
	(0.6629)	(0.9244)	(0.5624)	(0.3858)
GOLD BUZZ	0.0040	0.0171	−0.0530	−0.0185
	(0.0347)	(0.0271)	(0.0303)	(0.0112)
EPU	0.0203	−0.0002	−0.0227	−0.0029
	(0.0365)	(0.0175)	(0.0273)	(0.0072)
FED STATEMENTS	−0.1780	−0.1301	0.2143	−0.0468
	(0.2120)	(0.1214)	(0.1515)	(0.0771)
Interest Rate	−0.0337	−0.0395	−0.0862	0.0326
	(0.0935)	(0.0783)	(0.0775)	(0.0303)
R2	0.0181		0.0216	

Note: The table reports the results of the least square regressions with one breakpoint of the entity groups 6 and 7. The breaks are manually determined. Regime 1: 01/01/2017–12/31/2019. Regime 2: 01/01/2020–12/31/2021. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

We now turn to the estimations and individual variables in more detail. The bitcoin positions of the three small entities display a strong persistence since the first lag of the change in the corresponding position is positive and significant for all periods. Given that we use daily data, this suggests that on the aggregate small entities build up positions gradually, which is intuitive since a larger number of investors is involved in the smaller entity categories. It may indicate an increasing small investor base and/or existing small investors adding bitcoin to their holdings over time. There is no such pattern for the medium and large entities since the corresponding estimates are insignificant for the former and negative for the latter.

Public information and media narratives about Bitcoin and gold also have differing effects on small and medium-sized entities. The first three entities react positively to the extent of media attention of Bitcoin and gold reflected in the buzz index. For entities holding 0.1 – 1 bitcoin, the effect for Bitcoin buzz can only be observed during the first observation period while gold buzz is only significant during the second sub-period for the smallest group. The results suggest that retail investors buy bitcoin during periods of high buzz or hype, regardless of the direction of the corresponding sentiment. We find a negative relationship to Bitcoin buzz for those holding 1 – 100 bitcoin and 100 – 10k bitcoin. The latter group, however, only shows a negative significant effect during the first period. The two largest entities do not react to Bitcoin or gold buzz at all.

The reaction of entities to price changes is rather mixed and depends on the analysed period. While entity group **1** does not significantly adjust their holdings in reaction to price changes at all, entity group **2** decreases its exposure to bitcoin when prices go up in period two, whereas an opposite effect is observed in the first period. Entity group **3** and **4** also increase their holdings in period one and decrease their holdings in period two when positive price changes are observed. The different signs probably reflect the fact that the effects of prices depend on previous changes and the current price level. Given the complexity of price signals, we have conducted additional estimates where we include price signals in the form of moving averages. These results are discussed further below and suggest that small-scale investors in the first entity react positively to price changes.

When discussing the role of sentiment, it is interesting to recall the results of Corbet et al. (2020a). According to their results, negative macroeconomic news tend to increase bitcoin returns. To link our findings with these results, we have analysed the relationship between the sentiment related to Bitcoin and the sentiment related to US stock markets. Both series are positively correlated with a coefficient around 0.17, illustrating that positive news about US stock markets coincide with positive news about Bitcoin. The corresponding buzz variables are also positively related. This suggests that media attention (on average) does not reflect the belief that negative stock market news are positive news for bitcoin.

The coefficient estimates for the Bitcoin sentiment in Table 8 and Table 9 are positive for the entity groups **1** and negative for the entity groups **2** and **4**. This suggests that positive news about Bitcoin (which coincides with positive news about stock markets) can also lead to a decrease, rather than an increase, in retail investor engagement. The result that investors entering the market via the smallest entity react positively to sentiment is intuitive. The negative effect for the other two entities could reflect the belief that bitcoin returns will increase following negative news. Overall, the results show that retails investors do pay attention to media attention (given the positive effects for buzz) without necessarily following the narrative embedded in the resulting sentiment.

Table 11
Breakpoint regressions with multiple breaks, ΔEntity Group 1.

	ΔEntity Group 1 Size 0.00–0.01 Bitcoin				
Regime	01/01/2017–12/28/2017	12/29/2017–09/29/2018	09/30/2018–07/11/2019	07/12/2019–03/20/2021	03/21/2021–12/31/2021
Observations	362	275	285	618	286
ΔEntity Group (−1)	**0.2372*****	**0.4707*****	**0.2196*****	0.0343	**0.3553*****
	(0.0644)	(0.0973)	(0.0810)	(0.0517)	(0.0566)
ΔBTC Price	−0.0006	**−0.0041****	0.0006	0.0003	−0.0008
	(0.0013)	(0.0018)	(0.0009)	(0.0005)	(0.0006)
BTC SENTIMENT	**0.2219*****	0.2329	0.0555	**0.0874****	−0.0220
	(0.0855)	(0.1444)	(0.0404)	(0.0345)	(0.0382)
BTC BUZZ	**0.0245*****	0.0010	**0.0032*****	**0.0094*****	**0.0014***
	(0.0039)	(0.0037)	(0.0012)	(0.0010)	(0.0008)
GOLD SENTIMENT	0.0408	0.1315	−0.0177	**0.0638****	−0.0328
	(0.1243)	(0.1024)	(0.0454)	(0.0251)	(0.0459)
GOLD BUZZ	0.0076	**0.0193*****	**0.0057*****	**0.0031*****	**0.0136*****
	(0.0057)	(0.0072)	(0.0019)	(0.0011)	(0.0023)
EPU	−0.0011	**0.0120****	**−0.0036****	−0.0002	**0.0042****
	(0.0080)	(0.0052)	(0.0019)	(0.0006)	(0.0019)
FED STATEMENTS	−0.0121	−0.0093	−0.0068	0.0149	−0.0159
	(0.0138)	(0.0126)	(0.0095)	(0.0113)	(0.0119)
Interest Rate	**−0.0738***	0.0314	**0.0914*****	−0.0045	−0.2469
	(0.0380)	(0.0301)	(0.0195)	(0.0029)	(0.1597)
R2	0.5560				

Note: The table reports the results of the least square regressions with multiple breakpoints of the entity group 1. Breaks are determined by Bai–Perron Tests. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

The control variables provide further interesting insights and confirm our results that small investors are more responsive to public information than large investors. The bitcoin holdings of the first two entities decrease in case of higher policy uncertainty during the second period but increase for the fifth entity in the second sub-period. These results align with previous results, which show that bitcoin can be characterized by hedge or safe haven properties. Fed statements hardly play a role. Solely the entity groups **2**, **4** and **5** react to Fed statements. Entity groups **1, 2** and **3** also react to interest rates by reducing their holdings, while group **5** react positively to interest rate increases in period one. This again illustrates that public information, in this case concerning financial markets in general, is predominantly relevant for small and partly medium entities.

Our results show that media attention and sentiment indicators are more significant for investors with smaller investment sizes. Retail investors appear to be more responsive to incoming information than large ones regarding price dynamics and uncertainty. The slightly stronger effect of sentiment and media attention compared to prices might be explained by the fact that the underlying media coverage tends to be less volatile and channels different information. From a more general view, these results show that the behaviour of small investors is largely driven by public information, while large-scale investors do not react to such information, implying that their behaviour is driven by private information or specific expectations which are not directly observable. One possible explanation is that large investors often react opposite to market trends compared to small investors due to such private information. Another possible explanation is that these differences may be due to differing portfolio considerations of retail and large investors. As previously illustrated, all entities show significant variation but changes in large portfolios are more likely to be affected by (long-run) strategic decisions, which are detached from the short-run information our empirical approach captures.

To see whether the selected breakpoints drive our results, we have conducted further estimations where breakpoints have endogenously been determined, as mentioned earlier. The results of these regressions can be found in Table 11 – Table 14. These regressions confirm the picture we have obtained from the previous estimates concerning the explanatory power of our model and the role of buzz and policy uncertainty in explaining the bitcoin holdings of entities. As outlined above, no regressor besides the own lag is significant for the two largest entities. The results confirm that the smallest group of investors responds positively to higher sentiment about Bitcoin. The results also confirm that the smaller entities are much more responsive to the independent variables since the response to the incoming information varies over the different estimation periods. This is evident because the small entity groups exhibit multiple breakpoints, which progressively decrease with increasing entity size. Entity group **1** shows four breakpoints while entity groups **2** and **3** have three breakpoints. Entity group **4** has two breakpoints. The largest groups number **5, 6** and **7** do not exhibit any breakpoints. As explained earlier, particularly for the two largest entity groups, this might be due to the possibility that we are mainly observing exchanges and not individuals.

5.2. Robustness test and extensions

We have performed several additional robustness checks and extensions related to our control variables, the price and sentiment signals, the inclusion of nonlinearities and the construction of the entities. We have included the tables of robustness checks and extensions in the appendix to this paper, as we will only briefly discuss the following results and quickly highlight particular findings.

Table 12
Breakpoint regressions with multiple breaks, ΔEntity Group 2.

	ΔEntity Group 2 Size 0.01–0.1 Bitcoin			
Regime	01/01/2017–12/09/2017	12/10/2017–03/26/2019	03/27/2019–12/03/2020	12/04/2020–02/31/2021
Observations	343	472	618	393
ΔEntity Group (−1)	0.0524	**0.3947*****	0.0044	**0.2065*****
	(0.0471)	(0.0735)	(0.0501)	(0.0526)
ΔBTC Price	**0.0037***	−0.0007	−0.0014	**−0.0015****
	(0.0020)	(0.0010)	(0.0010)	(0.0006)
BTC SENTIMENT	0.1688	0.1332	**−0.1326*****	**−0.1055*****
	(0.1590)	(0.0880)	(0.0443)	(0.0357)
BTC BUZZ	**0.0414*****	**0.0108*****	**0.0073*****	**0.0033*****
	(0.0051)	(0.0034)	(0.0014)	(0.0008)
GOLD SENTIMENT	**0.4918*****	−0.0389	−0.0460	−0.0100
	(0.1845)	(0.0674)	(0.0395)	(0.0568)
GOLD BUZZ	**0.0739*****	0.0035	**0.0068*****	**0.0138*****
	(0.0114)	(0.0044)	(0.0015)	(0.0023)
EPU	0.0149	0.0006	0.0002	**0.0026***
	(0.0098)	(0.0030)	(0.0009)	(0.0015)
FED STATEMENTS	**−0.0443****	−0.0205	0.0111	**0.0300***
	(0.0205)	(0.0159)	(0.0118)	(0.0158)
Interest Rate	0.0049	−0.0089	**−0.0066***	−0.2026
	(0.0555)	(0.0108)	(0.0038)	(0.1739)
R2		0.4502		

Note: The table reports the results of the least square regressions with multiple breakpoints of the entity group 2. Breaks are determined by Bai–Perron Tests. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.
* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

Table 13
Breakpoint regressions with multiple breaks, ΔEntity Group 3.

	ΔEntity Group 3 Size 0.1–1 Bitcoin			
Regime	01/01/2017–12/19/2017	12/20/2017–11/25/2018	11/26/2018–07/17/2020	07/18/2020–12/31/2021
Observations	353	341	600	532
ΔEntity Group (−1)	**0.1425****	**−0.4060*****	**0.1783*****	**0.2611*****
	(0.0575)	(0.1388)	(0.0410)	(0.0554)
ΔBTC Price	**0.0082*****	0.0022	**−0.0018***	**0.0017*****
	(0.0019)	(0.0022)	(0.0011)	(0.0005)
BTC SENTIMENT	−0.2103	−0.0751	**−0.1942*****	−0.0114
	(0.1711)	(0.1381)	(0.0411)	(0.0320)
BTC BUZZ	**0.0233*****	**−0.0165*****	**0.0027***	−0.0005
	(0.0038)	(0.0056)	(0.0015)	(0.0009)
GOLD SENTIMENT	**0.4821****	0.0400	−0.0581	−0.0197
	(0.1993)	(0.1280)	(0.0385)	(0.0378)
GOLD BUZZ	**0.0908*****	**0.0246*****	**0.0067*****	**0.004*****
	(0.0124)	(0.0078)	(0.0016)	(0.0015)
EPU	**0.0259****	**−0.0109***	0.0014*	**−0.0039*****
	(0.0120)	(0.0064)	(0.0008)	(0.0011)
FED STATEMENTS	0.0106	0.0002	0.0095	0.0024
	(0.0363)	(0.0287)	(0.0076)	(0.0103)
Interest Rate	0.0662	**−0.1081*****	0.0026	0.0896
	(0.0551)	(0.0274)	(0.0033)	(0.1305)
R2		0.4218		

Note: The table reports the results of the least square regressions with multiple breakpoints of the entity group 3. Breaks are determined by Bai–Perron Tests. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.
* $p < 0.1$.
** $p < 0.05$.
*** $p < 0.01$.

Investors may not only consider recent price and sentiment signals but may also focus on the underlying trends. We, therefore, have modified the sentiment and price variable in the regressions with a 28-day moving average, allowing for the possibility that investors track recent trends in sentiment over time and not individual data points. Table 15 – Table 17 provide the corresponding results. While, in some instances, the significance and direction of the coefficients change, the overall picture is quite similar to that of our main regressions. Here also, smaller entities react more strongly to public information than the bigger ones and when

Table 14
Breakpoint regressions with multiple breaks, ΔEntity Groups 4, 5, 6 and 7.

	ΔEntity Group 4 1–100 BTC			ΔEntity Group 5 100–10k BTC	ΔEntity Group 6 10k–100k BTC	ΔEntity Group 7 >100k BTC
Regime	01/01/2017–24/11/2017	11/25/2017–11/25/2021	11/26/2018–12/31/2021	01/01/2017–12/31/2021	01/01/2017–12/31/2021	01/01/2017–12/31/2021
Observations	328	366	1132	1826	1826	1826
ΔEntity Group (−1)	0.1294	−0.0339	−0.0595	−0.0221	**−0.1156*****	**−0.0997****
	(0.1185)	(0.0715)	(0.0654)	(0.0208)	(0.0382)	(0.0438)
ΔBTC Price	**0.0072*****	**0.0029****	−0.0008	−0.0005	0.0057	−0.0075
	(0.0020)	(0.0014)	(0.0006)	(0.0011)	(0.0086)	(0.0062)
BTC SENTIMENT	**−0.2114***	**0.3638*****	**−0.1082*****	−0.0512	0.5553	0.0071
	(0.1136)	(0.1096)	(0.0365)	(0.0667)	(0.5762)	(0.3689)
BTC BUZZ	−0.0027	**−0.0095*****	**−0.0052*****	**−0.0028***	0.0070	0.0088
	(0.0069)	(0.0027)	(0.0011)	(0.0016)	(0.0155)	(0.0099)
GOLD SENTIMENT	**0.3520****	0.0494	0.0170	0.0648	−0.0567	−0.1732
	(0.1411)	(0.0725)	(0.0399)	(0.0675)	(0.5263)	(0.3678)
GOLD BUZZ	**0.0308*****	**0.0141*****	**0.0047*****	0.0037	0.0113	**−0.0324****
	(0.0080)	(0.0050)	(0.0015)	(0.0027)	(0.0211)	(0.0139)
EPU	0.0127	0.0023	0.0003	**0.0030****	0.0033	−0.0101
	(0.0080)	(0.0045)	(0.0006)	(0.0013)	(0.0130)	(0.0073)
FED STATEMENTS	−0.0454	−0.0226	−0.0048	−0.0240	−0.1728	0.1469
	(0.0352)	(0.0168)	(0.0128)	(0.0251)	(0.1488)	(0.1062)
Interest Rate	**0.1157*****	**−0.0669*****	**0.0092*****	0.0020	−0.0130	−0.0007
	(0.0443)	(0.0167)	(0.0025)	(0.0059)	(0.0429)	(0.0205)
R2		0.1347		0.0051	0.0159	0.0159

Note: The table reports the results of the least square regressions with multiple breakpoints of the entity groups 4, 5, 6 and 7. Breaks are determined by Bai–Perron Tests. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

looking at Bitcoin sentiment, the buzz variable continues to be the most important one. Further, we have estimated the regressions with the split sentiment variables (positive and negative sentiment) (see Table 18 – Table 20). We, however, find the split Bitcoin sentiment indices to be mostly insignificant, not adding further insights compared to the combined sentiment estimations.

Besides these robustness tests, we have conducted several additional estimates, which are available upon request. First, we included the change in the gold price and the VIX index as additional regressors. We ran both single break and multiple break regressions. The inclusion of these variables does not change the results significantly. However, the number of observations available drops considerably. Therefore, we stick to the original choice of variables.

To allow for a different form of nonlinearity, we have also experimented with smooth transition and threshold models to detect whether specific values of sentiment or buzz trigger nonlinear effects as transition variables. This could be the case, for example, if the impact of sentiment or price is greater during periods of higher sentiment or media attention. Others, such as (Entrop et al., 2020) or (Ben-Rephael et al., 2017), also only take extreme sentiment values into account and construct a dummy that takes the value one if the observation is above the median of the previous 20 periods and zero otherwise. To retain more information, we have kept the individual values of the 10% most extreme sentiment values. However, the results did not provide clear patterns, illustrating that the breakpoint approach is more adequate. We have also changed the potential link between sentiment and investor behaviour by including sentiment and other regressors with a lag of one period. The underlying idea is to allow for the possibility that investors take a day to adjust their portfolios.

Finally, we have analysed disaggregated entities instead of aggregating specific subsets as outlined earlier. This provides a more granular perspective on the aggregated entities but again leaves our main results concerning the distinction between small and large entities unchanged. We, therefore, have stuck with the main specification, which enables us to present the results in a more concise way.

6. Summary and concluding remarks

This paper has shed some light on what drives decision-making by investors in the bitcoin market. Based on the unique data set by Glassnode, which distinguishes between different investor groups by investment size, we investigated the determinants of investor positions in the bitcoin market conditional on the size of their positions.

We find that the reaction of market participants to price and sentiment changes depends on the size of the respective bitcoin portfolio by investors. Both explanatory power and coefficient estimates suggest that small-size investors react strongly to incoming public information and media narratives, with their decision-making strongly affected by sentiment and media attention. On the opposite, the response of large-scale investors to public information is much weaker, suggesting that their behaviour is driven by private information.

We also find that the directional response differs between both groups. Small-scale investors tend to increase their position when the bitcoin price increases and sentiment and buzz are high. In contrast, large-scale investors can act in the opposite direction. An

explanation is that large-scale investors often have different information and experience compared to small-scale investors. Moreover, they apply different investment strategies and work with different, presumably larger, time horizons.

Our results offer several avenues for future research. One existing gap in the literature corresponds to a deeper understanding of individual behaviour in the crypto market. Survey-based evidence on investor behaviour on the Bitcoin market would nicely complement our results but is difficult to obtain due to data availability reasons, particularly for large investors. The work by Weber et al. (2023) offers some first insights into the role of cryptocurrencies in household finance. Further representative surveys could at least shed some light on the motives of market participants entering the bitcoin market. In the same spirit, disaggregated data on large-scale investors could also be useful to assess whether large positions have specific effects on market volatility and sentiment. In this context, the relevance of media narratives on investor behaviour in the bitcoin market could be analysed at a more granular level.

CRediT authorship contribution statement

Joscha Beckmann: Writing – original draft, Supervision, Resources, Methodology, Formal analysis, Conceptualization. **Teo Geldner:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Jan Wüstenfeld:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization.

Data availability

The authors do not have permission to share data.

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All authors approved the version of the manuscript to be published.

Appendix

See Tables 15–20.

Table 15
28-days moving average breakpoint regressions with one break, ΔEntity Groups 1, 2 and 3.

	ΔEntity Group 1 Size 0.00–0.01 Bitcoin		ΔEntity Group 2 Size 0.01–0.1 Bitcoin		ΔEntity Group 3 Size 0.1–1 Bitcoin	
Regime	1	2	1	2	1	2
Observations	1068	731	1068	731	1068	731
ΔEntity Group (−1)	**0.4179*****	**0.3258*****	**0.3174*****	**0.2396*****	**0.2102*****	**0.2200*****
	(0.0738)	(0.0471)	(0.0527)	(0.0466)	(0.0664)	(0.0440)
ΔBTC Price	**0.0204*****	−0.0012	**0.0236*****	**−0.0093***	**0.0287*****	**−0.0112*****
	(0.0066)	(0.0038)	(0.0081)	(0.0048)	(0.0086)	(0.0041)
BTC SENTIMENT	0.2007	0.0314	−0.2226	0.1431	−0.2764	0.1238*
	(0.1692)	(0.0652)	(0.2180)	(0.0891)	(0.2164)	(0.0751)
BTC BUZZ	0.0038	0.0018	−0.0009	**0.0067*****	0.0017	**0.0043*****
	(0.0025)	(0.0016)	(0.0038)	(0.0019)	(0.0052)	(0.0013)
GOLD SENTIMENT	−0.1501	0.0336	0.1400	**−0.1848****	0.0955	**−0.3882*****
	(0.1179)	(0.0588)	(0.1825)	(0.0807)	(0.2058)	(0.0790)
GOLD BUZZ	**−0.0496*****	**−0.007****	−0.0119	−0.0054	0.0042	−0.0016
	(0.0086)	(0.0029)	(0.0111)	(0.0042)	(0.0143)	(0.0036)
EPU	0.0074	−0.0005	−0.0037	0.0002	**−0.0157***	0.0006
	(0.0046)	(0.0010)	(0.0081)	(0.0015)	(0.0090)	(0.0013)
FED STATEMENTS	0.2611	0.0251	0.0466	0.0199	0.0907	−0.0525
	(0.1774)	(0.0719)	(0.2333)	(0.0944)	(0.2564)	(0.0790)
Interest Rate	−0.0040	**−0.0156*****	−0.0161	**−0.0137*****	**−0.0439*****	−0.0029
	(0.0104)	(0.0027)	(0.0125)	(0.0038)	(0.0146)	(0.0038)
R2	0.4352		0.1814		0.1503	

Note: The table reports the results of the moving average least square regressions with one breakpoint of the entity groups 1, 2, and 3. The breaks are manually determined. The independent variables are a 28-days moving average. Regime 1: 01/28/2017–12/31/2019. Regime 2: 01/01/2020–12/31/2021. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

Table 16
28-days moving average breakpoint regressions with one break, ΔEntity Groups 4 and 5.

	ΔEntity Group 4 Size 1–100 Bitcoin		ΔEntity Group 5 Size 100–10k Bitcoin	
Regime	1	2	1	2
Observations	1068	731	1068	731
ΔEntity Group (−1)	0.0867	0.0184	−0.0261	−0.0298
	(0.0719)	(0.0591)	(0.0230)	(0.0429)
ΔBTC Price	**0.0112****	**−0.0110*****	−0.0026	−0.0086
	(0.0053)	(0.0040)	(0.0082)	(0.0104)
BTC SENTIMENT	**−0.3024***	−0.0015	−0.2078	**0.3653***
	(0.1600)	(0.0789)	(0.3726)	(0.2107)
BTC BUZZ	**−0.0056****	**−0.0035****	−0.0086	0.0061
	(0.0025)	(0.0014)	(0.0057)	(0.0041)
GOLD SENTIMENT	0.0105	**−0.2430*****	**0.8010***	0.0369
	(0.1405)	(0.0869)	(0.4664)	(0.1753)
GOLD BUZZ	0.0129	−0.0007	0.0127	0.0041
	(0.0108)	(0.0032)	(0.0168)	(0.0079)
EPU	−0.0002	0.0007	−0.0041	**0.0045***
	(0.0077)	(0.0012)	(0.0138)	(0.0028)
FED STATEMENTS	−0.0492	−0.1132	−0.2353	0.0661
	(0.1994)	(0.0857)	(0.2948)	(0.1960)
Interest Rate	0.0062	0.0000	0.0271	0.0065
	(0.0121)	(0.0065)	(0.0240)	(0.0102)
R2	0.0348		0.0121	

Note: The table reports the results of the moving average least square regressions with one breakpoint of the entity groups 4 and 5. The breaks are manually determined. The independent variables are a 28-days moving average. Regime 1: 01/28/2017–12/31/2019. Regime 2: 01/01/2020–12/31/2021. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

Table 17
28-days moving average breakpoint regressions with one break, ΔEntity Groups 6 and 7.

	ΔEntity Group 6 Size 10k–100k Bitcoin		ΔEntity Group 7 Size >100k Bitcoin	
Regime	1	2	1	2
Observations	1068	731	1068	731
ΔEntity Group (−1)	**−0.1266*****	**−0.1072***	**−0.1174****	−0.0169
	(0.0469)	(0.0612)	(0.0525)	(0.0664)
ΔBTC Price	0.0147	0.0984	−0.0331	0.0023
	(0.0589)	(0.1134)	(0.0484)	(0.0470)
BTC SENTIMENT	0.4349	−3.4729	1.2522	0.6441
	(2.0033)	(2.1333)	(1.5422)	(0.8903)
BTC BUZZ	−0.0073	−0.0475	0.0401	0.0120
	(0.0369)	(0.0407)	(0.0274)	(0.0168)
GOLD SENTIMENT	−1.7497	0.5074	−0.7956	0.0874
	(2.1140)	(2.1206)	(1.3169)	(0.8517)
GOLD BUZZ	−0.0647	−0.0218	−0.0238	−0.0039
	(0.0972)	(0.0566)	(0.0779)	(0.0236)
EPU	0.0070	−0.0191	0.0068	−0.0006
	(0.0613)	(0.0269)	(0.0437)	(0.0115)
FED STATEMENTS	−0.5935	2.9568	1.4452	−1.1054
	(2.0579)	(2.3523)	(1.7495)	(0.9159)
Interest Rate	0.1046	−0.0595	**−0.2010****	0.0294
	(0.1458)	(0.0873)	(0.1008)	(0.0355)
R2	0.0180		0.0174	

Note: The table reports the results of the moving average least square regressions with one breakpoint of the entity groups 6 and 7. The breaks are manually determined. The independent variables are a 28-days moving average. Regime 1: 01/28/2017–12/31/2019. Regime 2: 01/01/2020–12/31/2021. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

Table 18
Positive–negative sentiment breakpoint regressions with one break, ΔEntity Groups 1, 2 and 3.

	ΔEntity Group 1 Size 0.00–0.01 Bitcoin		ΔEntity Group 2 Size 0.01–0.1 Bitcoin		ΔEntity Group 3 Size 0.1–1 Bitcoin	
Regime	1	2	1	2	1	2
Observations	1095	731	1095	731	1095	731
ΔEntity Group (−1)	**0.4735*****	**0.1761*****	**0.2757*****	**0.1001****	**0.1847*****	**0.3046*****
	(0.0631)	(0.0424)	(0.0460)	(0.0440)	(0.0626)	(0.0422)
ΔBTC Price	−0.0006	−0.0003	**0.0026***	**−0.0017****	**0.0056*****	**−0.0022*****
	(0.0010)	(0.0005)	(0.0016)	(0.0007)	(0.0016)	(0.0009)
BTC SENT_Positive	−0.2002	−0.2292	−0.2797	**−0.3898****	**−0.9945****	−0.1507
	(0.1834)	(0.1422)	(0.3588)	(0.1558)	(0.3925)	(0.1387)
BTC SENT_Negative	**−0.5755*****	**−0.2666****	−0.3834	−0.1808	−0.7546	−0.0494
	(0.1821)	(0.1286)	(0.4037)	(0.1489)	(0.4610)	(0.1347)
BTC BUZZ	**0.0098*****	**0.0056*****	**0.0115*****	**0.0061*****	0.0054	−0.0002
	(0.0021)	(0.0007)	(0.0028)	(0.0008)	(0.0034)	(0.0009)
GOLD SENT_Positive	−0.0284	−0.0747	0.3131	−0.1043	0.3290	−0.1048
	(0.1192)	(0.0876)	(0.2252)	(0.1124)	(0.2360)	(0.0931)
GOLD SENT_Negative	0.0662	**−0.2255****	0.1878	−0.0393	0.1893	0.1334
	(0.1507)	(0.1014)	(0.2402)	(0.1256)	(0.2938)	(0.1178)
GOLD BUZZ	−0.0008	**0.0049*****	**0.0130*****	**0.0065*****	**0.0159*****	**0.0059*****
	(0.0024)	(0.0010)	(0.0039)	(0.0012)	(0.0046)	(0.0013)
EPU	−0.0009	**−0.0024*****	0.0000	**−0.0019*****	0.0007	−0.0003
	(0.0024)	(0.0005)	(0.0033)	(0.0006)	(0.0041)	(0.0006)
FED STATEMENTS	0.0025	0.0056	**−0.0251****	**0.0324****	0.0080	0.0135
	(0.0078)	(0.0086)	(0.0124)	(0.0107)	(0.0177)	(0.0103)
Interest Rate	**−0.0645*****	**−0.0161*****	**−0.0864*****	**−0.0209*****	**−0.0770*****	**−0.0090*****
	(0.0082)	(0.0026)	(0.0101)	(0.0039)	(0.0121)	(0.0036)
R2	0.4635		0.2700		0.1945	

Note: The table reports the results of the least square regressions with one breakpoint of the entity groups 1, 2 and 3. The breaks are manually determined. The sentiment are split in positive and negative. Regime 1: 01/01/2017–12/31/2019. Regime 2: 01/01/2020–12/31/2021. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

Table 19
Positive–negative sentiment breakpoint regressions with one break, ΔEntity Groups 4 and 5.

	ΔEntity Group 4 Size 1–100 Bitcoin		ΔEntity Group 5 Size 100–10k Bitcoin	
Regime	1	2	1	2
Observations	1068	731	1068	731
ΔEntity Group (−1)	0.0778	0.0024	−0.0297	−0.0165
	(0.0702)	(0.0613)	(0.0242)	(0.0423)
ΔBTC Price	**0.0043*****	**−0.0011***	−0.0009	0.0012
	(0.0011)	(0.0007)	(0.0015)	(0.0014)
BTC SENT_Positive	−0.2027	0.0376	−0.0990	0.2194
	(0.2719)	(0.1682)	(0.4052)	(0.3923)
BTC SENT_Negative	−0.0439	0.2053	0.1262	0.3271
	(0.2793)	(0.1587)	(0.4109)	(0.3565)
BTC BUZZ	**−0.0073*****	**−0.0053*****	**−0.0087*****	0.0009
	(0.0020)	(0.0012)	(0.0031)	(0.0020)
GOLD SENT_Positive	0.2310	**−0.2889*****	0.0970	0.2447
	(0.1620)	(0.1095)	(0.2391)	(0.2304)
GOLD SENT_Negative	−0.0357	**−0.3038****	−0.1016	0.1750
	(0.1767)	(0.1452)	(0.2925)	(0.2991)
GOLD BUZZ	**0.0126*****	**0.0045*****	0.0071	−0.0001
	(0.0029)	(0.0015)	(0.0052)	(0.0029)
EPU	0.0018	0.0000	0.0022	**0.0024***
	(0.0032)	(0.0006)	(0.0045)	(0.0013)
FED STATEMENTS	−0.0257	−0.0015	**−0.0581***	**0.0576***
	(0.0162)	(0.0121)	(0.0311)	(0.0325)
Interest Rate	0.0041	0.0018	**0.0269***	0.0011
	(0.0080)	(0.0066)	(0.0144)	(0.0106)
R2	0.0778		0.0154	

Note: The table reports the results of the least square regressions with one breakpoint of the entity groups 4 and 5. The breaks are manually determined. The sentiment are split in positive and negative. Regime 1: 01/01/2017–12/31/2019. Regime 2: 01/01/2020–12/31/2021. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

Table 20
Positive–negative sentiment breakpoint regressions with one break, ΔEntity Groups 6 and 7.

	ΔEntity Group 6 Size 10k–100k Bitcoin		ΔEntity Group 7 Size >100k Bitcoin	
Regime	1	2	1	2
Observations	1095	731	1095	731
ΔEntity Group (−1)	**−0.1290*****	**−0.1108***	**−0.1212****	−0.0221
	(0.0466)	(0.0625)	(0.0519)	(0.0670)
ΔBTC Price	0.0030	0.0099	−0.0119	−0.0015
	(0.0103)	(0.0170)	(0.0088)	(0.0071)
BTC SENT_Positive	0.3632	**−7.8982***	−0.2431	**2.8967***
	(2.7720)	(4.7778)	(2.5139)	(1.7566)
BTC SENT_Negative	0.3905	**−9.5772****	−1.2773	**3.2923****
	(2.9898)	(4.4280)	(2.8072)	(1.5918)
BTC BUZZ	0.0201	−0.0084	0.0206	0.0102
	(0.0229)	(0.0284)	(0.0183)	(0.0113)
GOLD SENT_Positive	1.3995	3.0200	−0.9964	−1.3667
	(1.7310)	(2.7388)	(1.5843)	(1.1455)
GOLD SENT_Negative	2.4642	3.0762	−0.5676	−1.4237
	(1.9283)	(3.2677)	(1.7376)	(1.3112)
GOLD BUZZ	−0.0076	0.0178	−0.0476	−0.0184
	(0.0344)	(0.0283)	(0.0298)	(0.0117)
EPU	0.0246	−0.0008	−0.0240	−0.0028
	(0.0359)	(0.0176)	(0.0271)	(0.0072)
FED STATEMENTS	−0.1727	−0.0716	0.2144	−0.0707
	(0.2116)	(0.1230)	(0.1525)	(0.0787)
Interest Rate	−0.0340	−0.0521	−0.0817	0.0391
	(0.0923)	(0.0822)	(0.0785)	(0.0311)
R2	0.0215		0.0230	

Note: The table reports the results of the least square regressions with one breakpoint of the entity groups 6 and 7. The breaks are manually determined. The sentiment are split in positive and negative. Regime 1: 01/01/2017–12/31/2019. Regime 2: 01/01/2020–12/31/2021. Significance is indicated by bold coefficients. Standard errors are heteroskedasticity-consistent and given in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

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